SKILL WALLET SMARTBRIDGE

Project Design Phase-II
Data Flow Diagram & User Stories

Date	01 November 2026
Team ID	[Enter Team ID]
Project Name	Global Olympic Sports Performance Analysis Using Tableau
Maximum Marks	4 Marks

Data Flow Diagrams:

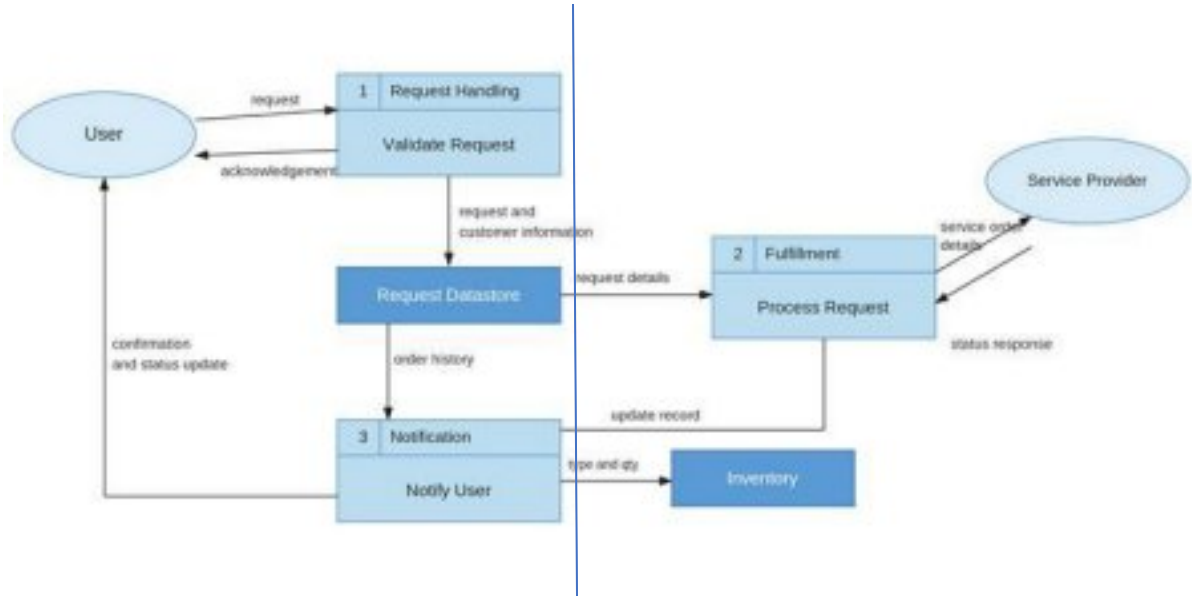
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).

Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example: DFD Level 0 (Industry Standard)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Project Analyst / Student	Data Import	USN-1	As the project analyst, I can connect the Olympic athlete_events.csv file to Tableau so that I can use the athlete, country, year, season, sport, event and medal data.	The CSV loads successfully and the expected fields and records are visible in Tableau.	High	Sprint-1
Project Analyst / Student	Data Preparation	USN-2	As the project analyst, I can check data types, missing values and inconsistent labels so that the analysis is reliable.	Year is numeric; dimensions have suitable data types; missing medal values are treated as non-medal records.	High	Sprint-1
Project Analyst / Student	Data Preparation	USN-3	As the project analyst, I can create a unique medal key so team-event athletes are not incorrectly counted as separate medals.	Distinct medal counts are used for country, sport and year comparisons; assumptions are documented.	High	Sprint-1
Priya – NOC Performance Strategist	Sport Performance	USN-4	As Priya, I can compare medals per athlete across sports so that I can identify strong sport categories.	A sorted horizontal bar chart displays the sport-wise measure with year and season filters.	High	Sprint-2
Priya – NOC Performance Strategist	Country Performance	USN-5	As Priya, I can compare medal totals across countries so that I can understand national performance.	A country ranking chart displays distinct medal totals and supports a country/year/ season selection.	High	Sprint-2
Priya – NOC Performance Strategist	Sport Distribution	USN-6	As Priya, I can view medal distribution by sport so that I can identify sports contributing most medals.	A treemap displays sport-wise distinct medal counts and responds to relevant filters.	High	Sprint-2

Carlos – Investment Analyst	Investment Analysis	USN-7	As Carlos, I can compare verified sports investment with medal outcomes so that I can assess funding effectiveness.	A scatter/bubble chart plots investment against medal outcomes by country; investment source, currency and year are documented.	High	Sprint-3
Carlos – Investment Analyst	Regional Analysis	USN-8	As Carlos, I can compare medal-efficiency measures by region so that I can identify regions for further investigation.	A regional chart displays a clearly defined efficiency calculation based on a region mapping table.	Medium	Sprint-3
Carlos – Investment Analyst	Country Efficiency	USN-9	As Carlos, I can rank countries by medal efficiency so that I can compare performance relative to athlete participation.	A sorted country chart displays the stated formula and allows year/season filtering.	Medium	Sprint-3
William Robert – Journalist	Historical Trends	USN-10	As William, I can view medal totals over Olympic years so that I can explain changes over time.	A line chart shows distinct medal totals by year and distinguishes Summer and Winter Games.	High	Sprint-4
William Robert – Journalist	Host Performance	USN-11	As William, I can compare each host nation's medal performance at its home Games so that I can explore the host advantage question.	A host comparison chart uses a verified year-season-host lookup and counts only medals earned by the host NOC at that edition.	High	Sprint-4
William Robert – Journalist	Regional Distribution	USN-12	As William, I can explore medal distribution by region and sport so that I can describe regional patterns.	A regional medal distribution chart supports sport filtering and uses a documented country-to-region mapping.	Medium	Sprint-4
Dashboard Viewer	Interactive Dashboards	USN-13	As a dashboard viewer, I can filter by year, season, country, sport and medal type so that I can explore the results relevant to me.	Filters update the applicable charts and tooltips show clear field definitions.	Medium	Sprint-5
Project Analyst / Student	Story & Reporting	USN-14	As the project analyst, I can present the three scenario dashboards as a Tableau story so that stakeholders can follow the analysis.	The story contains four points: project overview and one point for each scenario dashboard.	High	Sprint-5
Administrator / Project Analyst	Web Integration	USN-15	As an administrator, I can embed published Tableau dashboards in a Flask web page so that viewers can access the analysis through a simple interface.	The Flask page loads the authorized Tableau view and navigation links work; embedding is tested after publishing.	Medium	Sprint-6
Project Analyst / Student	Documentation	USN-16	As the project analyst, I can document the data preparation, calculations, dashboards and limitations	Documentation and end-to-end demonstration video describe	High	Sprint-6

Project DFD (Level 1 – textual representation):

External entities: Project Analyst / Student, Olympic Data Source, and Dashboard Viewer. Flow: Olympic CSV files + supplementary lookup files → (1) Connect & Import Data in Tableau → (2) Clean, validate and relate data → (3) Calculate medal KPIs and apply filters → (4) Build 9 visualizations → (5) Assemble 3 dashboards and 4-story-point narrative → (6) Publish/embed through Tableau and Flask (planned) → Dashboard Viewer. Data stores: Original CSV dataset; supplementary region/host/investment lookup files; Tableau workbook (.twb/.twbx). Data limitation: the supplied athlete_events.csv covers Olympic records through 2016 and does not contain sports-investment or region fields. These analyses require verified supplementary datasets; 2020–2026 claims must not be reported until corresponding records are added.